Logit and Probit

Logit and Probit Models are widely used in credit risk modeling to estimate the probability of a borrower defaulting on a loan. Both models are types of regression that help answer the fundamental question: *What is the probability that a borrower will default?*

Here's a detailed look into each model:

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1. Logit Model in Credit Risk

The **Logit model** (or logistic regression) is a statistical technique that models the probability of a binary outcome, such as default vs. non-default. The model uses the **logistic function** to ensure that probabilities stay between 0 and 1.

How the Logit Model Works

1. **Log-Odds Transformation**: The Logit model estimates the **log-odds** of default. The log-odds are a transformation that allows us to fit a linear model to probabilities:

$$\text{log-odds} = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

where:

- p is the probability of default,
- β_0 is the intercept, and
- $\beta_1, \beta_2, \ldots, \beta_n$ are coefficients for each predictor $x_1, x_2, \ldots, x_n$.

2. **Probability Calculation**: After estimating the log-odds, they are converted to probabilities using the logistic function:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)}}$$

This ensures that p lies between 0 and 1, making it a valid probability.

3. **Model Training**: The model is trained by finding the best values for the parameters $\beta_0, \beta_1, \ldots, \beta_n$ using a method like **Maximum Likelihood Estimation (MLE)**, which identifies the parameters that maximize the probability of observing the data.

4. **Interpretation of Coefficients**: In a logit model, each coefficient represents the change in the log-odds of default for a one-unit increase in the corresponding predictor variable. Larger coefficients indicate that the variable has a stronger impact on default likelihood.

Example

Let's say we're using two predictors, **Credit Score** and **Income**:

- A lower credit score might increase the likelihood of default (negative impact).
- Higher income might decrease the likelihood of default (positive impact).

The model might look like:

$$\text{log-odds} = -2 + 0.01 \cdot \text{Credit Score} - 0.0005 \cdot \text{Income}$$

2. Probit Model in Credit Risk

The **Probit model** is another type of regression used for binary outcomes, like predicting defaults. Unlike the Logit model, the Probit model assumes that the probability of default follows a **cumulative normal distribution**.

How the Probit Model Works

1. **Normal Distribution Assumption**: The Probit model assumes that there is a latent (hidden) variable that represents the propensity to default. This latent variable is normally distributed. If this variable crosses a certain threshold, the borrower is predicted to default.
2. **Probability Calculation**: The probability of default is calculated using the **cumulative distribution function (CDF)** of the normal distribution:

   $$p = \Phi(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)$$

where:

- Φ\Phi is the CDF of the standard normal distribution,
- β_0 is the intercept, and
- $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients for each predictor $x_1, x_2, \ldots, x_n$.

The normal CDF ensures that the probability, P, remains between 0 and 1.

1. **Model Training**: Like the Logit model, the Probit model's parameters are estimated using **Maximum Likelihood Estimation (MLE)**.
2. **Interpretation of Coefficients**: In a Probit model, each coefficient represents the change in the z-score (or standard deviations) associated with a one-unit change in the predictor variable. A positive coefficient means that an increase in that variable raises the likelihood of default.

Logit vs. Probit Model: Key Differences

Table 4

Feature	Logit Model	Probit Model
Probability Distribution	Logistic distribution (S-shaped)	Normal distribution (bell-shaped)
Function	Logistic function	Normal cumulative distribution function (CDF)
Interpretation	Log-odds	Z-scores (standard deviations)
Coefficients	Change in log-odds per unit change in predictor	Change in z-score per unit change in predictor
Tail Behavior	Slightly fatter tails	Normal distribution tails
Common Application	When odds ratios are of interest	When normal distribution assumptions are preferred

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Pros and Cons of Logit and Probit Models

Logit Model:

- **Pros**: Easier to interpret as odds ratios; computationally faster.
- **Cons**: Assumes a logistic distribution, which may not fit all data types.

Probit Model:

- **Pros**: Often a better fit when data aligns with a normal distribution; good for smaller samples.
- **Cons**: Less interpretable as odds ratios; computationally more complex.

Use Case: Sample Application in Credit Risk

Let's assume we want to build a Logit/Probit model to predict the probability of default based on features like **Credit Score** and **Debt-to-Income Ratio**.

Sample Code Using Python (Logit Model)

```
import pandas as pd
import numpy as np
import statsmodels.api as sm

# Sample dataset: Default data with predictors
data = pd.DataFrame({
    'credit_score': [650, 720, 580, 680, 700, 640],
    'debt_to_income_ratio': [0.4, 0.3, 0.6, 0.5, 0.3, 0.55],
    'default': [0, 0, 1, 0, 0, 1]  # 1 = default, 0 = no default
})

# Adding a constant for the intercept
data['intercept'] = 1

# Defining predictors and the target
X = data[['intercept', 'credit_score', 'debt_to_income_ratio']]
y = data['default']

# Logit model
logit_model = sm.Logit(y, X)
result = logit_model.fit()

# Output model summary
print(result.summary())

# Predict probability of default for a new borrower
new_borrower = pd.DataFrame({'intercept': [1], 'credit_score': [600], 'debt_to_income_ratio':
[0.5]})
predicted_prob = result.predict(new_borrower)
print(f"Predicted Probability of Default: {predicted_prob[0]:.2f}")
```

Interpretation

1. **Model Coefficients**: The output provides the coefficients for each predictor. A negative coefficient for `credit_score` suggests that a higher score reduces the likelihood of default. A positive coefficient for `debt_to_income_ratio` suggests that a higher ratio increases the default likelihood.
2. **Predicted Probability**: For a borrower with a credit score of 600 and a debt-to-income ratio of 0.5, the model might output a default probability of, say, 0.42, or 42%.

The main differences between the normal and logistic distributions are the shape of the tails and the behavior of the failure rate function:

- **Tails**
- The logistic distribution has longer tails than the normal distribution. This means that the logistic distribution represents a greater possibility of an event occurring further away from the mean.
- **Failure rate function**
- In the upper tail of the logistic distribution, the failure rate function levels out for large values, approaching 1/If location parameter.
- **Robustness**
- The logistic distribution's heavier tails often make analyses based on it more robust than those based on the normal distribution.